AI USAGE DISCLOSURE STATEMENT

(Software Engineering/MIS/Network Security Capstone Project)

University: Duy Tan University
Faculty / Program: International School – Software Engineering
Course: Capstone Project / Graduation Project

Project Title: Smart Elderly Care AI - A Multimodal AI System for Proactive Monitoring, Forecasting, and Care of Elderly Aged 60 and Above.

Team Name: C1SE.34

Mentor: PhD Dieu,Huynh Ba

Student ID	Full Name	Role in Project
29211155327	Hieu, Duong Van	Leader, Scrum Master, QA, Developer, Tester
29219022025	Nghia, Nguyen Huu	AI Developer, Backend Developer, Frontend Developer, Tester
29211155992	Duy, Tran Van	AI Developer, Backend Developer, Frontend Developer, Tester
29211157943	Vinh, Ngo Le	Backend Developer, Frontend Developer, Tester
29201155816	My, Phan Nguyen Giang	BA, UX/UI Designer, Frontend Developer, Backend Developer, Tester, Document

1. AI TOOLS DECLARATION

Please list all AI tools that may be used in this project.

AI Tool	Version / Platform	Intended Purpose
ChatGPT	GPT-5.6 Luna	**General Troubleshooting & Design Exploration:** Inquiring into software architecture patterns, explaining third-party library behaviors, sanity-checking code snippets, and drafting boilerplate REST endpoints
DeepSeek	V3	**Algorithmic Reasoning & Logic Optimization:** Analyzing mathematical logic for 17-keypoint skeleton pose estimation (spinal tilt angle and fall

AI Tool	Version / Platform	Intended Purpose
		trajectory calculations); conducting root-cause analysis for complex asynchronous concurrency (FastAPI, WebSockets, and multithreaded edge processing)
Gemini AI	Gemini 3.8 Flash	**Datasheet Synthesis & Academic Cross-referencing:** Leveraging its large context window to parse and synthesize extensive technical datasheets (Rockchip RK3588S NPU specifications, AMG8833 thermal matrix I2C protocol, Bleak BLE GATT stack); cross-referencing WHO/CDC clinical guidelines on elderly falls.
GitHub Copilot	VS Code Extension	**Code Scaffolding & Autocompletion:** Autocompleting repetitive boilerplate code .
Claude	Claude 3.5	**Engineering Documentation Refinement:** Auditing and polishing technical English syntax for academic deliverables (Proposal, SRS, SAD) according to IEEE standards; refactoring code modules to adhere to Clean Code conventions.

2. EXPECTED AI USAGE BY PROJECT PHASE

Indicate how AI tools may assist during development.

Phase	AI Tools	Main Purpose
Problem Analysis	Gemini AI, ChatGPT	Research analysis, summarization, and competitor benchmarking
System Design	DeepSeek, Gemini AI, Claude	Architecture, database, hardware-interface, and diagram validation
Implementation	DeepSeek, GitHub Copilot, ChatGPT	Edge AI, sensor fusion, model optimization, UI/API development
Debugging	DeepSeek, ChatGPT	Error diagnosis, memory/concurrency debugging, runtime troubleshooting
Testing	DeepSeek, Gemini AI, GitHub Copilot	Synthetic data generation and automated test development
Documentation	Gemini AI, Claude, ChatGPT	Consistency checking, technical writing, and documentation refinement

3. ESTIMATED EXTENT OF AI ASSISTANCE

Provide an estimated percentage of AI-assisted work.

Component	Estimated AI Assistance
Backend / Core Logic	20%
Frontend / UI	25%

Component	Estimated AI Assistance
Testing	25%
Documentation	30%

4. RESPONSIBLE AI USE CHECKLIST

Please confirm the following:

Statement	Yes
We will disclose all AI tools used in the project	☒
AI-generated code will be reviewed and understood by the team	☒
AI will be used only as a support tool, not as a full solution generator	☒
All team members can explain the system design and implementation	☒
We agree to demonstrate code ownership during the final defense	☒

5. ACADEMIC INTEGRITY DECLARATION

We acknowledge that:

- AI tools serve strictly as assistive technologies and do not substitute for engineering judgment, system design, or domain problem-solving.
- All AI-generated suggestions will be verified, modified, and validated against engineering constraints before integration.
- The team maintains sole accountability for system safety, data integrity, and operational reliability.
- Undisclosed or unverified deployment of AI-generated content constitutes academic misconduct under Duy Tan University regulations

6. TEAM DECLARATION

We confirm that the information provided above is accurate and that we will follow the Responsible AI Use Policy for Capstone Projects.

Name	Signature	Date
Hieu, Duong Van		
Duy, Tran Van		
My, Phan Nguyen Giang		
Nghia, Nguyen Huu		
Vinh, Ngo Le		

AI POLICY FOR SOFTWARE ENGINEERING CAPSTONE PROJECTS
(Responsible and Ethical Use of Artificial Intelligence in Capstone Development)

Institution: Duy Tan University

Faculty / School: International School
Program: Software Engineering / Management Information System / Network Security
Effective Date: Mar 15th, 2026

1. Purpose of the Policy

This policy establishes guidelines for the responsible, transparent, and ethical use of Artificial Intelligence (AI) tools in Software Engineering Capstone projects.

The objectives are to:

1. Encourage productive and responsible use of AI tools in software development.
2. Ensure academic integrity and originality of student work.
3. Prevent misuse of AI that results in superficial or unverified project outcomes.
4. Ensure that students demonstrate genuine technical competence and understanding.
5. Maintain the academic and professional standards expected from a university-level engineering capstone project.

AI tools are recognized as legitimate development aids; however, students remain fully responsible for the intellectual and technical content of their projects.

2. Scope

This policy applies to:

- All **Capstone / Senior Projects / Graduation Projects**
- All students enrolled in:
    - Software Engineering
    - Management Information Systems
    - Related computing programs
- All phases of the project lifecycle:
    - Proposal
    - Design
    - Implementation
    - Testing
    - Documentation
    - Final defense

3. Definition of AI Tools

For the purposes of this policy, AI tools include but are not limited to:

3.1 Generative AI

Examples:

- ChatGPT
- Claude
- Gemini
- LLaMA-based assistants

Uses may include:

- concept clarification
- documentation drafting
- code explanation
- debugging assistance

3.2 AI Coding Assistants

Examples:

- GitHub Copilot
- Amazon CodeWhisperer
- Tabnine
- Cursor AI

Uses may include:

- code suggestion
- autocomplete
- test generation

3.3 AI Design and Media Tools

Examples:

- Midjourney
- DALL·E
- Stable Diffusion
- Figma AI plugins

Uses may include:

- UI/UX concept generation
- visual prototyping

3.4 AI Data Analysis Tools

Examples:

- Python AI libraries
- AutoML tools
- LLM-based data analysis assistants

4. Acceptable Uses of AI

Students are allowed to use AI tools for supportive purposes, including:

4.1 Learning and Understanding

- explanation of programming concepts
- clarification of algorithms
- understanding frameworks and libraries

4.2 Development Assistance

AI may assist with:

- code suggestions
- debugging hints
- refactoring ideas
- generating test cases
- generating boilerplate code

However, students must:

- review
- understand
- modify

all AI-generated content.

4.3 Documentation Support

AI may assist in:

- drafting documentation
- formatting reports
- summarizing references

Students must edit and verify accuracy.

4.4 Brainstorming and Design Ideation

AI can be used to:

- explore architecture patterns
- compare technical solutions

- generate early design ideas

Final design decisions must be made and justified by the student team.

5. Unacceptable Uses of AI

The following uses are strictly prohibited:

5.1 AI-Generated Projects

Submitting a project where:

- the majority of code is generated by AI
- students cannot explain the system

5.2 Blind Code Copying

Using AI-generated code:

- without understanding
- without validation
- without modification

5.3 AI-Generated Documentation Without Verification

Submitting documentation that is:

- entirely AI-generated
- not reviewed by students

5.4 AI-Based Academic Misconduct

Examples include:

- generating full reports using AI
- generating design documents without team involvement
- fabricating research references
- submitting AI-generated work as original intellectual contribution

6. AI Usage Disclosure Requirement

All teams must disclose AI usage in their project documentation.

Disclosure must include:

1. AI tools used
2. purpose of usage
3. project phase where AI was applied

4. extent of AI-generated content

This disclosure must appear in:

- Proposal document
- Final report
- Appendix (AI Usage Statement)

Failure to disclose AI usage may be considered academic misconduct.

7. Student Responsibility

Students remain **fully responsible** for:

- system architecture
- algorithmic design
- code correctness
- testing results
- documentation accuracy

During project defense, students must be able to:

- explain their system architecture
- explain implementation decisions
- debug their system
- modify code when requested

8. Evaluation Criteria Related to AI Usage

Project evaluation will consider:

Criterion	Description
Technical Understanding	Students understand their own system
Original Engineering Work	Evidence of human design and reasoning
Responsible AI Use	AI used as tool, not replacement
Code Ownership	Students can explain and modify code
Documentation Integrity	Report reflects genuine work

If a team cannot explain their code, the project may receive significant penalties.

9. Verification Methods

Supervisors and evaluation committees may verify authenticity by:

- requesting live code explanation
- asking students to modify code during defense

- requesting design justification
- reviewing AI usage logs

Failure to demonstrate ownership may result in:

- grade reduction
- project rejection
- academic integrity investigation

10. Ethical Use of AI

Students must adhere to ethical standards including:

- transparency
- accountability
- intellectual honesty
- respect for intellectual property

AI must be used to enhance learning, not to replace engineering competence.

11. Policy Violations

Violations of this policy may result in:

- grade penalties
- project rejection
- academic integrity proceedings
- disciplinary action according to university regulations

12. Student Acknowledgement

All students must sign the following statement:

"We acknowledge that we have read and understood the AI Policy for Software Engineering Capstone Projects. We commit to using AI tools responsibly, transparently, and ethically, and we accept full responsibility for the technical and intellectual content of our project."

Name	**Signature**	**Date**
Hieu, Duong Van		
Duy, Tran Van		
My, Phan Nguyen Giang		
Nghia, Nguyen Huu		
Vinh, Ngo Le		

13. Supervisor Confirmation

Signature: ……………………………………… Date: ………………………………………